ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Robert Hooke	Hooke saw box-like compartments and called them cells.
2	**(b)** cell, tissue, organ, organ system, organism	Cells form tissues; tissues form organs, and so on.
3	**(b)** vacuole	A large vacuole stores water and substances.
4	**(c)** microscope	A microscope magnifies tiny things.
5	**(c)** tissue	A tissue is a group of similar cells.
6	**(d)** give shape and protection to plant cells	It is rigid and strong.
7	**(c)** nucleus	The nucleus holds the genetic material and directs cell activities.
8	**(b)** an organ	It is made up of several tissues working together.
9	**(d)** a simple microscope with a single lens	He ground his own lenses.
10	**(c)** cell membrane	It is selectively permeable.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Robert Hooke in 1665 (1); He observed thin slices of cork under a microscope (1).	1+1
12	It contains the genetic material (DNA) (1); It controls all activities of the cell (1).	1+1
13	Bacteria lack a membrane-bound nucleus (1); Their genetic material lies free in the cytoplasm (nucleoid) (1).	1+1
14	Group of similar cells doing a particular function (1); e.g. muscle tissue or xylem (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Peel a thin layer and place it in water on a slide (1); Add a drop of stain and cover with a cover slip carefully (1); Observe under the microscope and draw the rectangular cells (1).	1+1+1
16	All living bodies are made of cells (1); Each cell performs life functions (1); New cells arise from existing cells (1).	1+1+1
17	Cell wall is outside, rigid, in plants only (1); Cell membrane is thin and flexible, in all cells (1); Membrane controls what enters and leaves the cell (1).	1+1+1
	OR: Plant cell has a cell wall; animal cell does not (1); Plant cell has chloroplasts; animal cell does not (1); Plant cells have a large vacuole; animal cells have small ones (1).	1+1+1
18	Yeast: bread making (1); Bacteria: curd formation from milk (1); Fungi (e.g. moulds): production of antibiotics like penicillin (1).	1+1+1
	OR: Chloroplasts are found in plant cells (1); They contain chlorophyll to trap sunlight (1); Chlorophyll gives green colour to leaves (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cell wall: shape and protection (1); Cell membrane: controls entry/exit (1); Cytoplasm: site of cell activities (1); Nucleus: controls cell and has genetic material (1); Chloroplast and large vacuole: food making; storage (1).	1+1+1+1+1
	OR: Bacteria: single-celled, no nucleus (1); Viruses: non-cellular, parasites (1); Fungi: lack chlorophyll; yeast, mushroom (1); Protozoa: single-celled, with nucleus; Amoeba (1); Each can cause disease in some cases (1).	1+1+1+1+1
20	Cell: e.g. muscle cell (1); Tissue: group of muscle cells (1); Organ: heart (1); Organ system: circulatory system (1); Organism: the whole body (1).	1+1+1+1+1
	OR: Nerve cell: long to transmit messages (1); Red blood cell: disc-shaped for carrying oxygen (1); Muscle cell: elongated to contract (1); Root hair cell: long to absorb water (1); Structure fits function in all cells (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that animal cells also have a cell wall or chloroplasts.
- Confusing the **cell wall** (outer, rigid, plants) with the **cell membrane** (inside the wall, in all cells).
- Believing cork contains living cells, when Hooke saw the walls of dead cells.

- Calculating total magnification by adding the lens powers instead of multiplying.
- Writing that all cells have a nucleus; bacteria have no true nucleus.
- Forgetting the stain, thin peel and cover slip when describing slide preparation.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** multicellular	Many cells make the body.
2	**(b)** their functions differ	Shape fits function.
3	**(d)** cell wall	Plant cells have a rigid outer wall.
4	**(c)** well-defined nucleus	Their genetic material lies free in the cytoplasm.
5	**(a)** carry messages over long distances	Structure suits function.
6	**(c)** chloroplast	Chlorophyll in chloroplasts traps light for photosynthesis.
7	**(b)** Paramecium	Paramecium has a single cell.
8	**(b)** carrying oxygen	They carry oxygen with haemoglobin.
9	**(c)** stain makes the cell parts easier to see	Stain gives contrast.
10	**(c)** unicellular fungus	It is used in baking and brewing.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cell parts are almost transparent (1); Stain gives contrast so parts become visible (1).	1+1
12	Cell, tissue, organ (1); Organ system, organism (1).	1+1
13	Cell wall (1); Chloroplast (and large central vacuole) (1).	1+1
14	Shape is related to the function of the cell (1); e.g. long nerve cells carry messages (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	It magnifies objects too small for the eye (1); Cells, bacteria and protozoa can be seen (1); It led to the cell theory and understanding of diseases (1).	1+1+1
16	They are single-celled (1); No well-defined nucleus (1); They are found everywhere and some cause diseases (1).	1+1+1
17	Bacteria: Lactobacillus (1); Protozoa: Amoeba (1); Fungi: yeast (1).	1+1+1
	OR: Cell membrane: controls movement of substances (1); Cytoplasm: site of cell activities (1); Nucleus: controls the cell (1).	1+1+1
18	Viruses are non-cellular; bacteria are cells (1); Viruses multiply only inside host cells (1); Antibiotics work on bacteria but not on viruses (1).	1+1+1
	OR: A single fertilised cell divides again and again (1); Cells get specialised (1); They form tissues, organs and organ systems (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hooke observed cork and named cells (1665) (1); Leeuwenhoek made single-lens microscopes (1); He saw bacteria and protozoa in pond water, teeth scrapings (1); These observations opened the study of microscopic life (1); Later scientists framed the cell theory (1).	1+1+1+1+1
	OR: Has an eyepiece lens and objective lenses (1); Light passes through the specimen and lens (1); Image is magnified twice (1); Total = eyepiece x objective (1); 10x and 40x gives 400x (1).	1+1+1+1+1
20	Unicellular: single cell body (1); One cell does all functions (1); Multicellular: many cells; division of labour (1); e.g. Amoeba, yeast (1); e.g. humans, trees (1).	1+1+1+1+1
	OR: No chlorophyll, so cannot make food (1); Have cell walls (1); Feed on dead or living organisms (1); Yeast: baking and brewing (1); Mushrooms: food (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that animal cells also have a cell wall or chloroplasts.
- Confusing the **cell wall** (outer, rigid, plants) with the **cell membrane** (inside the wall, in all cells).
- Believing cork contains living cells, when Hooke saw the walls of dead cells.

- Calculating total magnification by adding the lens powers instead of multiplying.
- Writing that all cells have a nucleus; bacteria have no true nucleus.
- Forgetting the stain, thin peel and cover slip when describing slide preparation.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** bacterial cell	Bacteria are microscopic.
2	**(b)** cytoplasm	Cytoplasm holds the organelles.
3	**(d)** bacteria	Bacteria are typically a few micrometres across.
4	**(c)** photosynthesis	They trap sunlight to make food.
5	**(a)** unicellular organism	The whole body is one cell.
6	**(a)** virus	Viruses multiply only inside host cells.
7	**(b)** pathogens	Pathogens cause diseases.
8	**(b)** no cell wall	Animal cells lack a cell wall and chloroplasts.
9	**(b)** fungi	Fungi lack chlorophyll.
10	**(c)** cell	All living things are made of cells.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They do not have a cellular structure (1); They multiply only inside a host cell (1).	1+1
12	Unicellular: one cell, e.g. Amoeba (1); Multicellular: many cells, e.g. a mango tree (1).	1+1
13	Encloses the cell contents (1); Controls entry and exit of substances (1).	1+1
14	Used in baking bread (1); Used in making alcoholic drinks / fermentation (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Tissue: group of similar cells, e.g. muscle (1); Organ: tissues working together, e.g. stomach (1); An organ is at a higher level of organisation (1).	1+1+1
16	Nerve cells are long and branched (1); Red blood cells are disc-shaped and small (1); Muscle cells are long and contractile (1).	1+1+1
17	Carry it with both hands (1); Do not touch the lens with fingers (1); Focus slowly and clean the lens gently (1).	1+1+1
	OR: Their whole body is a single cell (1); The single cell takes food, respires and excretes (1); It also reproduces (1).	1+1+1
18	They help in making curd, bread, vinegar (1); Decomposers recycle dead matter (1); They are used to make medicines and enzymes (1).	1+1+1
	OR: Magnification is the number of times an image is enlarged (1); Total = 10 x 40 (1); = 400x (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Scrape inner cheek gently with a clean toothpick (1); Spread on a slide with a drop of water (1); Stain with methylene blue or safranin (1); Cover with a cover slip and observe (1); Rounded, irregular cells with a stained nucleus and no cell wall (1).	1+1+1+1+1
	OR: All living things are made of one or more cells (1); Life processes such as nutrition and reproduction occur in cells (1); Cells group into tissues, organs and organisms (1); Hooke named the cell in 1665 (1); Cells differ in shape and size according to their function (1).	1+1+1+1+1
20	Single-celled, tiny, no true nucleus (1); Shapes: rods, spheres and spirals (1); Useful: curd, decomposition, nitrogen fixing (1); Harmful: cholera, typhoid, tuberculosis (1); Can be seen only under a microscope (1).	1+1+1+1+1
	OR: Harmful: cause diseases like cholera (1); Harmful: spoil food (1); Beneficial: curd formation (1); Beneficial: decomposers recycle matter (1); Antibiotics and vaccines come from microbes (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that animal cells also have a cell wall or chloroplasts.
- Confusing the **cell wall** (outer, rigid, plants) with the **cell membrane** (inside the wall, in all cells).
- Believing cork contains living cells, when Hooke saw the walls of dead cells.

- Calculating total magnification by adding the lens powers instead of multiplying.
- Writing that all cells have a nucleus; bacteria have no true nucleus.
- Forgetting the stain, thin peel and cover slip when describing slide preparation.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND